

ImaGen: AI Image Generator in Android and IOS platform

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This Report Presented in Partial Fulfillment of the Requirements for the
Degree of Bachelor of Science in Computer Science and Engineering

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APPROVAL

This Project/internship titled “**IMAGEN: AI IMAGE GENERATOR IN ANDROID AND IOS PLATFORM**”, submitted by **MD. Rafi Hasan AND Md. Mohtasim Billah**, ID No: **191-15-12837 AND 191-15-12577** to the Department of Computer Science and Engineering, Daffodil International University has been accepted as satisfactory for the partial fulfilment of the requirements for the degree of B.Sc. in Computer Science and Engineering and approved as to its style and contents. The presentation has been held on January 2023.

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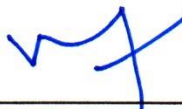


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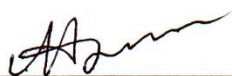
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DECLARATION

We thus certify that we worked on this under the direction of **Nazmun Nessa Moon**, **Associate Professor, CSE Department**, Daffodil International University. We further declare that no portion of this project or any other portion has been submitted elsewhere for consideration of any degree or diploma.

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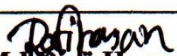
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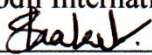


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ABSTRACT

This project “ImaGen: AI Image Generator in Android and IOS platform” is a system for getting high-quality and realistic images from text. Many apps are providing this service. But most of the apps are not able to generate realistic and high-resolution images according to the text. Which we are able to implement in our app. our app is used in various fields, including advertising, social media post, architecture, fashion, film, etc. We have built our app by following artificial intelligence techniques and combining some models. because AI is in the process of analyzing existing art and creating new art. In this project, we combine two models for the problem of text-conditional image generation. one is the CLIP model another is the diffusion model. The CLIP model is a successful representation learner for images, and diffusion models are generative modeling frameworks that have achieved state-of-the-image performance in image generation tasks. To develop this project, we have used Flutter Framework, Dart, as a programming language, and Visual Studio, Firebase, and an emulator for testing purposes. This app was tested in many ways after implementation and it works successfully.

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CHAPTER 1

INTRODUCTION

1.1 Introduction

As an artist draws with talent, we have taught our model how to generate images from text. Our own model can generate images with different filters from a text, our goal is not only to generate images. Our main goal is to get the image that a user is looking for with high resolution. An artist needs a lot of time to create images, which we have implemented in our app to find images quickly through search. our model allows you to generate images from scratch based on a text description. Essentially, you only need to input text describing an image you're thinking of, and the model will generate a surprisingly accurate image matching your description. It's especially useful for industries that rely heavily on visual information, advertising, social media post, architecture, fashion, and film etc. It enables the production of wholly original artwork that is unrestricted by the preconceptions and limitations of human artists.

1.2 Motivation

Nowadays we need different types of images which we create through an artist. It takes a lot of time to process the image. The present time is the age of internet, everything is available on the internet now. Considering the time of a user, we have implemented this project through an app, so that a user can easily get their desired image. Many websites, apps that provide images but most of them are paid version. In this place, we will provide free images to users to make their daily life easier. We have seen many places where a user fails to find the image he is looking for. We have this app. The main motive is to give them the desired image.

1.3 Objectives

- Produce High-Quality Image
- generate realistic images from text.

- faster result basis on search.
- Creating image-realistic 3D renderings.
- Generating images for using various purposes.
- Creating new art or visualizations.
- The users generate a larger number of images in a shorter amount of time.
- To save time and money.

1.4 Expected Outcome

1. We turn an artist's thinking into digital art.
2. The arts are divided into different Filter so that the user can search according to the filter.
3. People will not need to go to the artist physically.
4. Getting information through the app is possible using the search option.
5. It will save their time, complexities, money and other things.
6. Users can easily download their desired images.

1.5 Project Management and Finance

Given the short amount of time, we'll need to make a schedule in order to do the work by the due date. It also applies to talking about a project that has a due date and needs to be finished right away.

Gantt Chart

Activities	W1	W2	W3	W4	W5	W6	W7	W8	W9	W10	W11	W12
Research												
Specification												
Planning												
Design												
Development												
Testing												
Assessment												
Documentation												

Table 1.5.1: Gantt Chart

Release plan and milestone

Here are the release schedule and milestones:

Activities	Duration in week	Total week
Research	W1, W2	2
Specifications	W2, W3	2
Planning	W3, W4	2
Design	W4, W5	2
Development	W5, W6, W7, W8	4
Testing	W8, W9	2
Assessment	W10, W11	2
Documentation	W11, W12	2
Software release	W12	1

Table 1.5.2: Release plan and milestone

1.6 Report Layout

Chapter 1: Introduction

This chapter has covered the project's motivation, goals, and anticipated results. The final section of this chapter covered the report layout.

Chapter 2: Background

This page discusses the project's historical context. Here, the extent of the issue, comparisons with other websites, App related work, and project hurdles are all described.

Chapter 3: Requirement Specification

This chapter discusses about the requirements. Business process modeling, the requirement collection and analysis, the use case model of the project with description, Activity Diagram, Flowchart diagram and the design requirements are the topics of this chapter.

Chapter 4: Design Specification

The project's designs are all included in this chapter. These include the implementation requirements, front-end design, back-end design, interaction design, and user experience.

Chapter 5: Implementation and Testing

This chapter's topics include database implementation, front-end designs and interactions, testing implementation, test results, and reporting.

Chapter 6: Conclusion and Future Scope

Conclusion and the project's future scope, goal, and Limitations are covered in the final chapter.

CHAPTER 2

BACKGROUND

2.1 Preliminaries/Terminologies

These assumptions might involve complex architectures, auxiliary losses, or side information such as object part labels or segmentation masks supplied during training [5]. we propose a novel photo-realistic text-to-image generation model that implicitly disentangles semantics to both fulfill the high-level semantic consistency and low-level semantic diversity [7]. We considered this issue in our study and built a deep learning-based architecture for semantically consistent image generation: recurrent convolutional generative adversarial network (RC-GAN). RC-GAN successfully bridges the advancements in text and picture modelling, converting visual notions from words to pixels [9]. Contrastive models like CLIP have been shown to learn robust representations of images that capture both semantics and style. To leverage these representations for image generation [11]. Text-to-image generation has traditionally focused on finding better modeling assumptions for training on a fixed dataset.

2.2 Related Work

Here are some of the related websites, and app which provides an ImaGen: AI Image Generator in Android and IOS platform. Most of them have limited features such as low-resolution images, unrealistic images, software performance issues, paid versions, etc. Some platforms for finding ImaGen: AI Image Generator in Android and IOS platforms are:

- Fotor [16]
- Nightcafe [17]
- Hotpot.ai [18]
- Stablecog [19]
- Lensa [20]
- Jasper art [21]

Features:

- huge amount of image data to produce images.
- Customization image that they want.
- Some has limited features for users.

Problems:

- Most of the software can't produce high quality image
- Can't produce unique and realistic image
- Search result are not faster.

2.3 Comparative Analysis And Summary

In our project have many features. Some are similar and some different comparing to the mentioned web sites and app. Moreover, we have developed it with Artificial intelligence technique which is more effective result and user friendly to users.

- What an artist imagines in drawing, to bring out his drawing, we can use artificial intelligence to make that drawing a reality, which is making our time more effective.
- We have implemented through this project how to get a unique and realistic image from a text. This process is very fast. We have focused on that so that the user can get the image very easily.
- Sometimes the image that the user expects does not get the full result, in that case, he can customize the image and make it according to his mind.
- Sometimes user fail to get high resolution image we will provide high resolution image to user through this project so that they can use it in their real life.
- Most of the images here are free version. An user can easily download these free version images.

2.4 Scope of the problem

I have faced many problems to complete this project. But I have overcome these difficulties. First of all, we found out how to create a digital image from a text. We used artificial intelligence to perform this process. we combined CLIP (Contrastive Language-Image Pre-training) and diffusion model to complete this project. It's best model to training the text to image generate. In our project, we have used flutter, dart, python etc. visual studio is the editor. Using flutter, users can run the software Android and IOS platform. As this app is based on user friendly, I hope many people will use this type of app which will reduce their complexities.

2.5 Challenges

Here is some challenges which I have faced are mentioned below:

- Building the model and connecting it to the app was very difficult.
- It took a lot of time to get the output and a lot of work was required to get the output faster.
- Device related issues were the biggest challenge for us at the beginning.
- To create a beautiful design and unique features.
- To developing a user friendly interface.

CHAPTER 3

REQUIREMENT SPECIFICATION

3.1 Development Model

To create the "ImaGen: AI Image Generator in Android and IOS platform" project, we used the agile software development technique. Agile software development is a framework that is used in software development projects but has no physical presence. The agile development method aids developers in optimizing productivity since the project is divided into smaller iterations. Due to its emphasis on people, the agile development strategy is currently the approach most frequently employed when building software.



3.2 Requirement Collection & Analysis

We must gather information outlining the starting work for every project that is in development. To create a graphical representation of the project, we need the specifications. Without a graphical view, a project is challenging to implement. we had a lot of difficulties gathering the necessary. For us, it was a huge challenge. It was more difficult for me to get information about requirements because we were working with team and it was difficult to find requirements everywhere. we must thus get knowledge from reliable sources. we also need to be extremely careful when gathering requirements. For our ImaGen: AI Image Generator in Android and IOS platform project. we gathered all the requirements before we began to construct my project. To use this project, we need to gather certain artificial intelligence tools. we must then gather the best method and model to use for this project. Find ways to improve the project's output results. we gathered requirements and examined them. Following investigation, we had many method models ready to describe the system. A successful project requires the collection of requirements, analysis of the acquired needs, and specification of the project's methodology.

3.3 Use Case Modeling and Description

For the development of any project, use case diagrams are crucial. A use case diagram shows the system from a high perspective. This diagram can be used to distinguish between internal and external variables. If anyone wants to use the system for the first time, then they must input information as email and password. after completing registration, users can search the image. users can search different filter to find image. The previous searches images are shown History page.

I have drawn one use case diagram for this project. Here is a user's initial use case diagram. where Figure 3.1 displays to us Use case model for user.

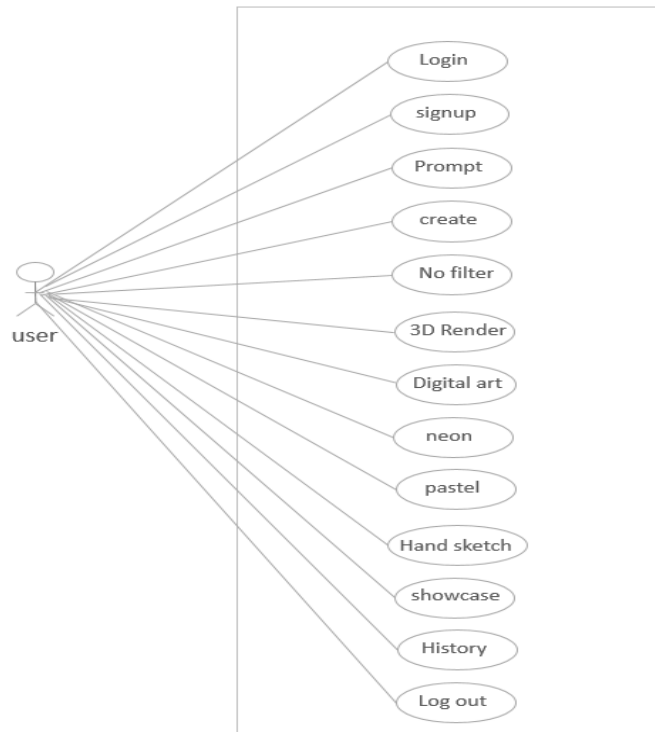


Figure 3.1: Use Case Model for Users

When the users will access our app, they have to must create an account. The users click the search box for finding the images, then click the create button to generate image based on the text. They will see different type of images. There are different filters in our app, 3D, no filter, neon, digital art, pastel, hand sketch etc. if user want, they can search any filter for images. There is a history page. in this page, the users will see previous searches images.

3.4 Activity Diagram

Registration

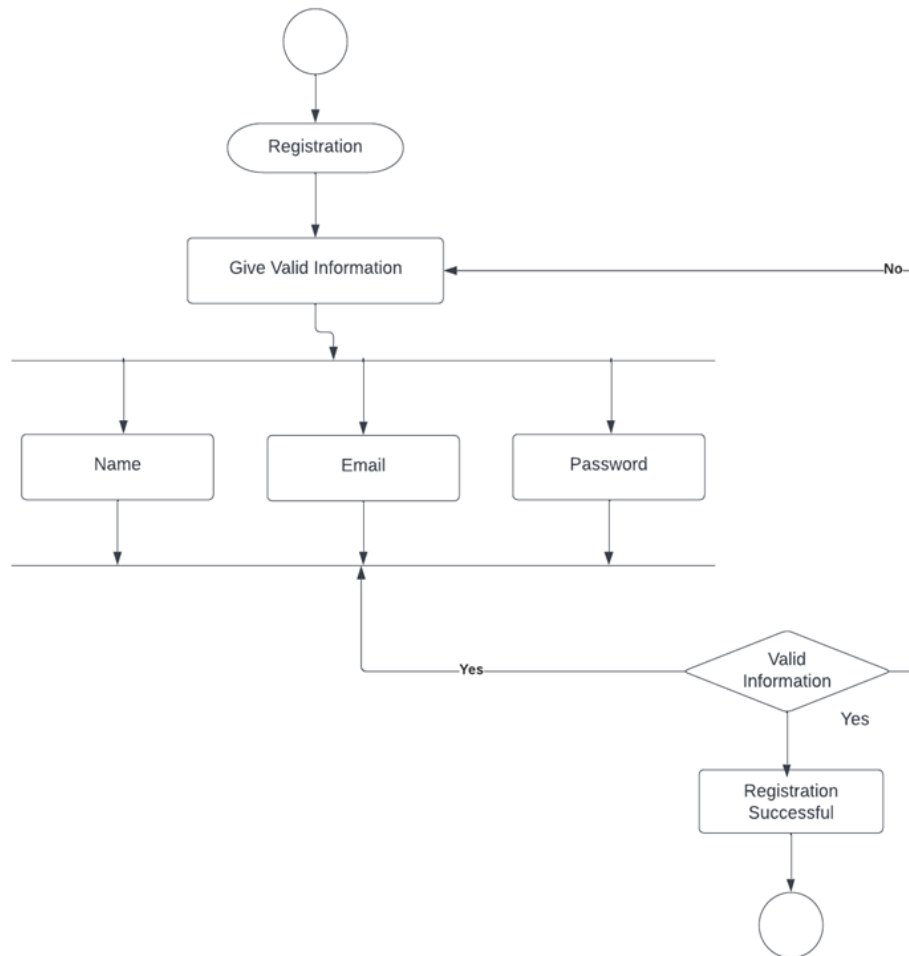


Figure 3.2: Activity Diagram of registration

This is the registration process in our app. first of all, the users have to fill up three input fields name, email, password with valid information. When the user will give the valid information, registration successfully message is shown our system.

Login

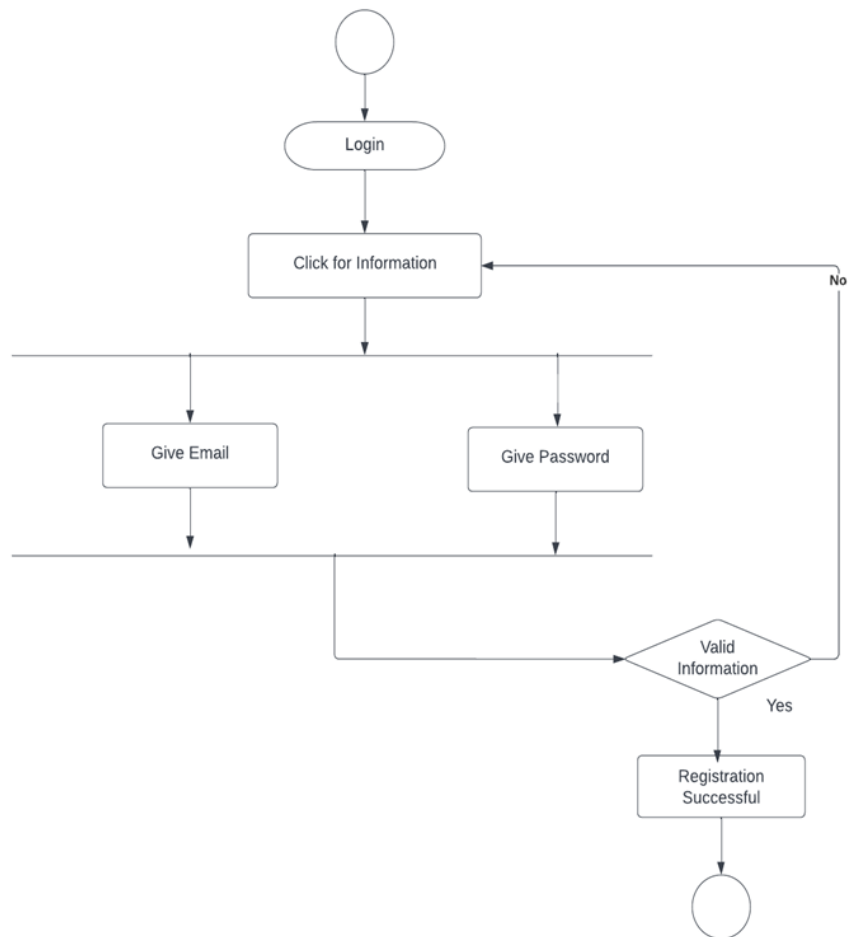


Figure 3.3: Activity Diagram of Login

This is the login process in our app. first of all, the users have to fill up two input fields email, password with valid information. When the user will give the valid information, Login successfully message is shown our system.

Create image

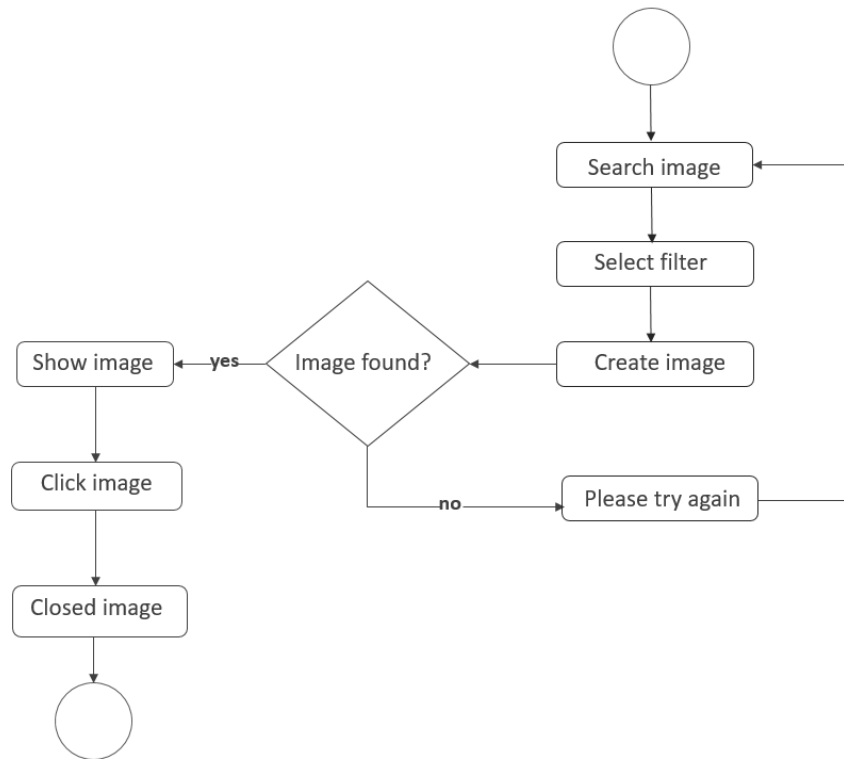


Figure 3.4: Activity Diagram of create image

First of all, users click in the search box for finding the image. When the text description will be finished, they can choose the filters, then press the create button to create different type of images. if the images are found, user will have shown different type of images. Then they can click any image and closed the image.

History

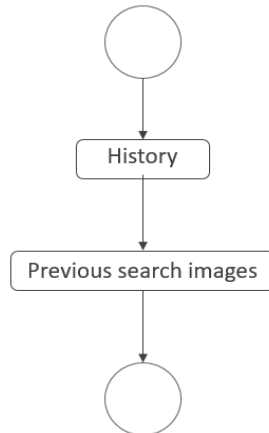


Figure 3.5: Activity Diagram of History

Click the history page, the history page will be shown the previous images.

Logout

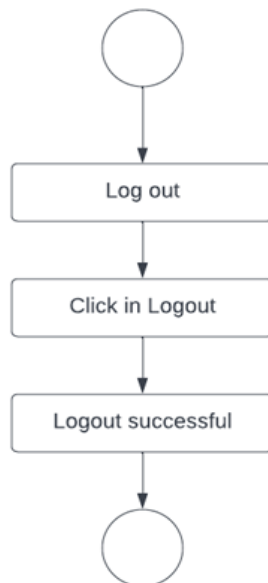


Figure 3.6: Activity Diagram of Logout

Click the logout button. Logout successfully

3.5 Flow Chart Diagram

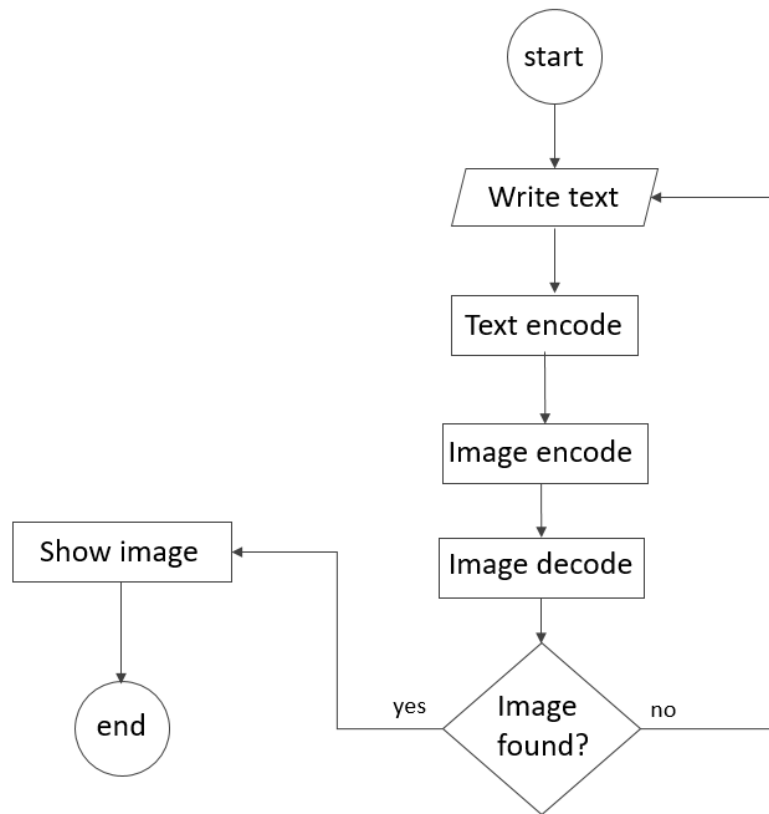


Figure 3.7: Flow chart diagram of the system

3.6 Design Requirements

- In this system has one user module, user module.
- General users have registration or sign up option.
- After sign in, users can show their profile information.
- User show the search button to search image.
- User show the create button to create the image.
- User show the Filtered of image.
- User show the search History.

CHAPTER 4

DESIGN SPECIFICATION

4.1 Front-End Design

In any application, front-end-design is very important because users interact with the application by this. We've tried to make it user friendly and easily understandable by the users. To make the user interface, we have used dart (programming-language). Our app has some pages, we needed to design every single page. Let's see some pages with their design.

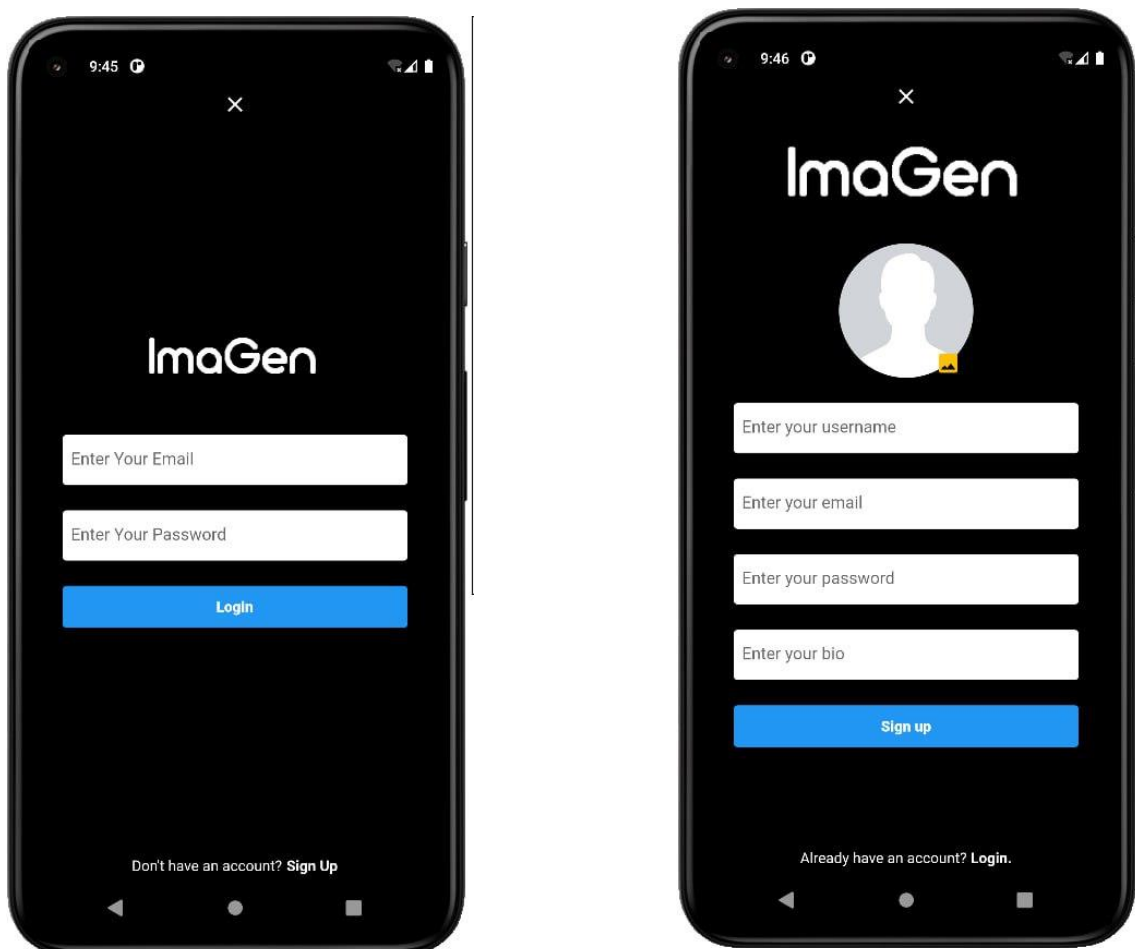


Figure 4.1: Login and Registration page

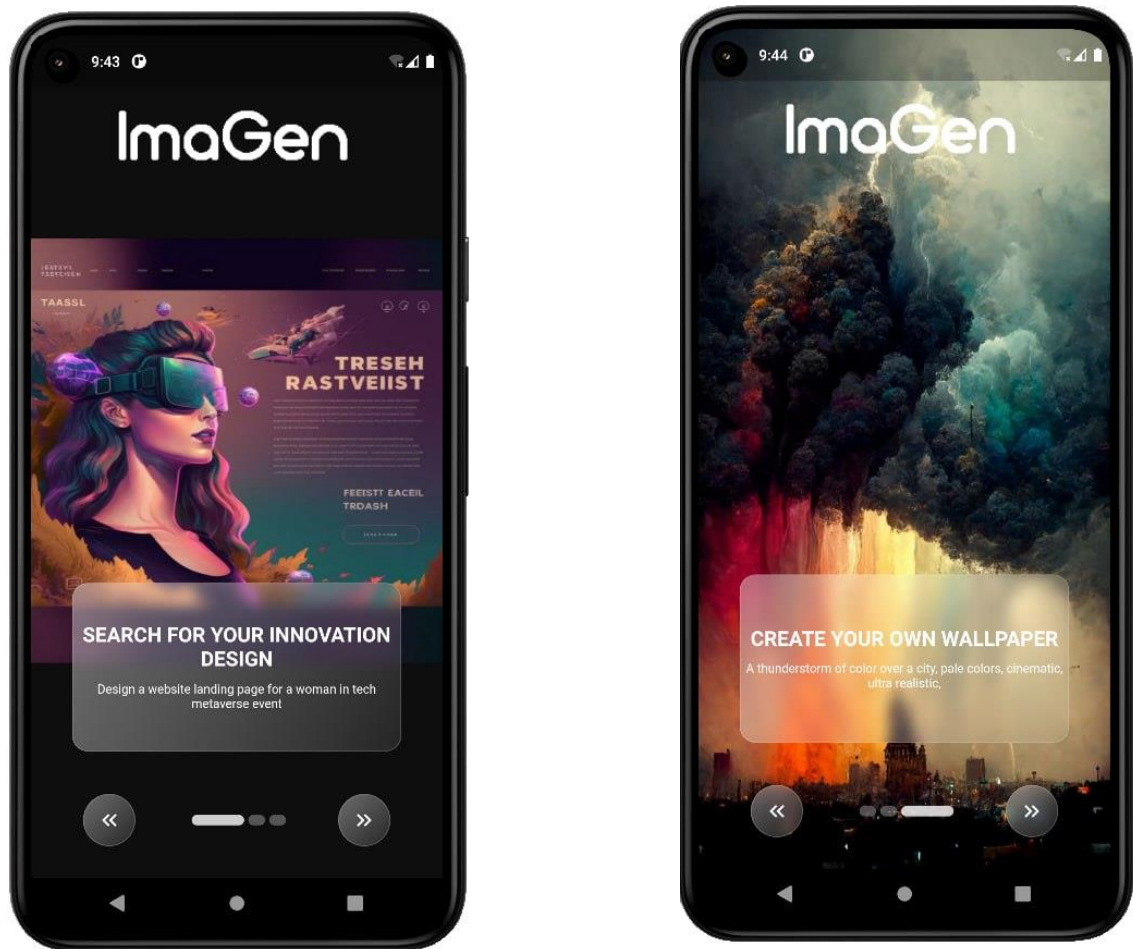


Figure 4.2: App slider

In Figure 4.2 is the app slider page. It is design with dart programming Language.

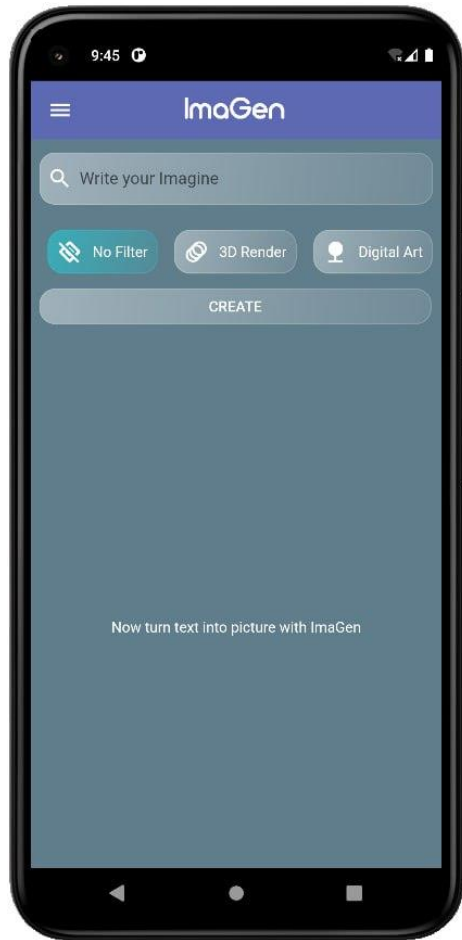


Figure 4.3: search and Filter page for finding images

This is the design of our search and Filter page for finding images. Here, in the figure 4.3

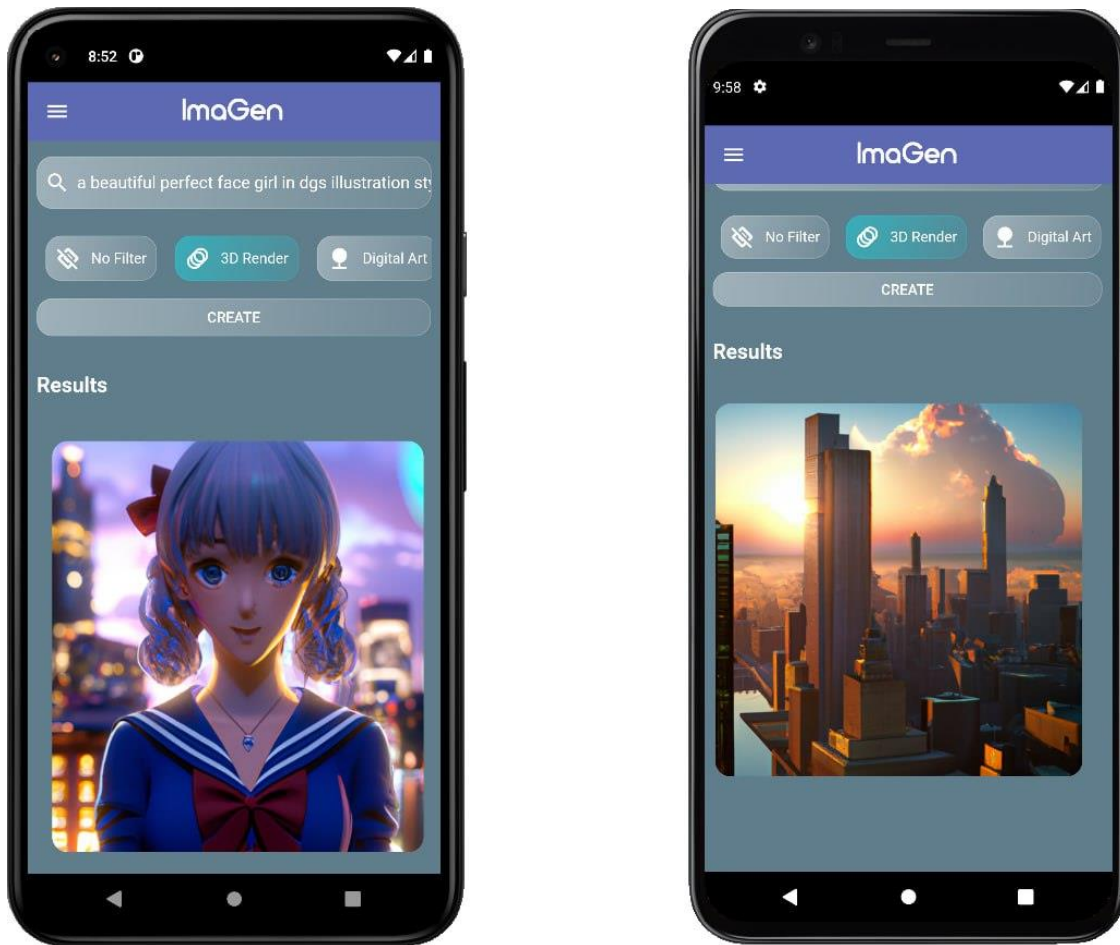


Figure 4.4: output result based on text.

4.2 Back-End Design

We have turned the customized model into an API. It's are deep learning models to generate digital images from natural language descriptions, called "prompts". Here we used "dio" to call the API. dio means an effective HTTP client for Dart that supports interceptors, global configuration, form data, request cancellation, file downloading, timeouts, and other features. Here we use dio to call the API. We have the prompt and effect by calling the API, then we get our desired result through API Get.

4.3 Interaction Design and UX

User Home Page Design:

- The user will see a search box to search for the image.
- There are six types of filter buttons, which means you can select their images according to the filter.
- There is a Create button through which the image will be created.
- There is a history button through which users can find their searched images.

4.4 Implementation Requirements

Requirements for the implementation of our project are discussed below:

- In our project, we used Flutter. The Dart programming language may be used to create native interfaces for IOS and Android utilizing the Google SDK Flutter. The Dart programming language, which is geared for creating quick programs on any platform, powers Flutter.
- We used Artificial intelligence in our project. using Ai tool, CLIP model and diffusion model.
- Emulator, a feature in visual studio, is a virtual operating system which can be created in visual studio to debug, test and run an application on different app devices of visual studio.

CHAPTER 5

IMPLEMENTATION AND TESTING

5.1 Implementation of the project:

Prompt Design:

Our models have the ability to write original stories and conduct complex text analyses. They can do so many various things, so you have to be exact in what you want.

Creating UI Design:

We implemented this project with flutter framework. We used Dart as programming language. The reason we chose flutter framework is that it is cross platform, it means it supports Android and IOS operating system.

Translator:

Our search box can be searched in four languages. The languages are English, Spanish, Arabic and Chinese. flutter has a package called translator, through which we can convert any language into English.

Provider:

We have used providers to improve the performance of the app. Many times the app has lazy loading and takes time to enter from one page to another. To avoid these things, we use providers. Also we see that in a project we need to create the same class in many places so we can use it in different places by calling using provider.

Hive:

In our project, we used Hive. because it's a lightweight, lightning-fast key-value database developed entirely in Dart that enables offline application data synchronization and storage. Hive, a key-value data store written in Dart, covers both simple and advanced data structures and delivers the best performance. It has some features such as cross platform, super performance, strong encryption built in, and no native dependencies etc.

CLIP (Contrastive Language-Image Pre-training)

To determine how much a particular text passage corresponds to an image, CLIP is trained on hundreds of millions of photos and the captions that go with them. In other words, CLIP just learns how closely connected any given caption is to a particular image rather than attempting to predict a caption given an image. CLIP can understand the relationship between textual and visual representations of the same abstract item thanks to its contrastive rather than predictive purpose.

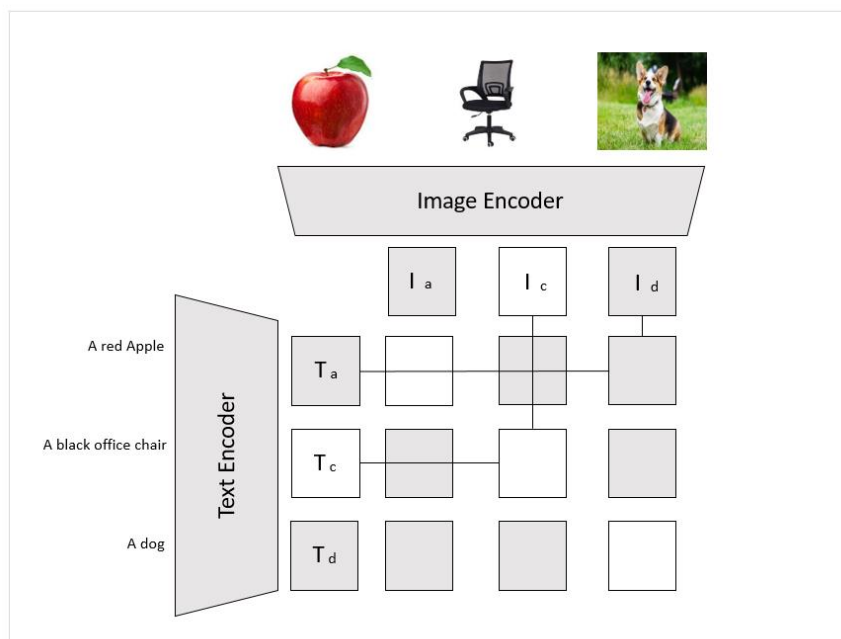


Figure 5.1: Figure of clip model

Diffusion Model

Diffusion models have proven to be highly effective at influencing image formation, such as text-guided image synthesis. Such achievement motivates several attempts to use pre-trained diffusion models on a broad scale to address the difficult issue of real image modification. Studies in this field discover a special textual token that corresponds to numerous photographs of the same thing. The Girl with a Pearl Earring painting is one example of a situation where there is just one image available. When pre-trained diffusion models are fine-tuned using just one image, there are serious overfitting problems.

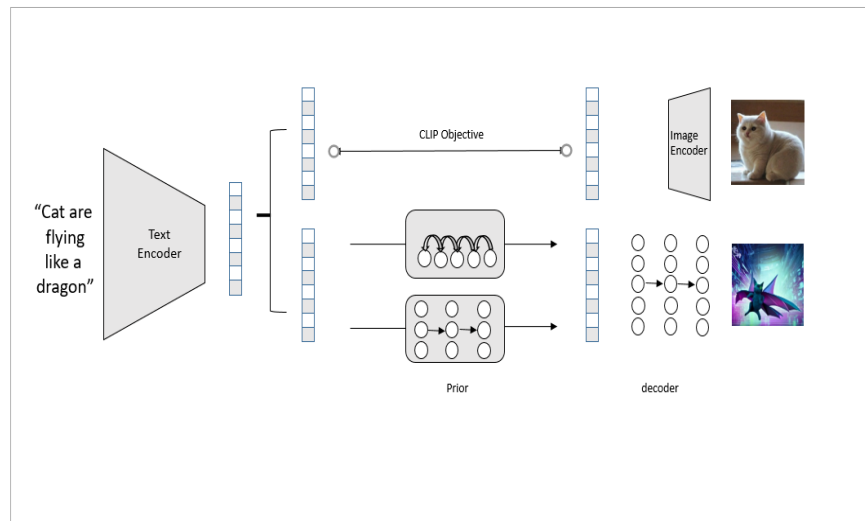


Figure 5.2: Figure of diffusion model

5.2 Implementation of Front-End Design

The layer or component of an application that the user can use, view, and interact with through buttons, graphics, interactive elements, navigational menus, and text is known as the front-end application, also referred to as the application's interface. Above we explained how we implemented our Ui design. now we showed six categories of image based on prompt.

Text: a lone skyscraper landscape vista photography by Carr Clifton & Galen Rowell, 16K resolution, Landscape veduta photo by Dustin Lefevre & tdraw, 8k resolution

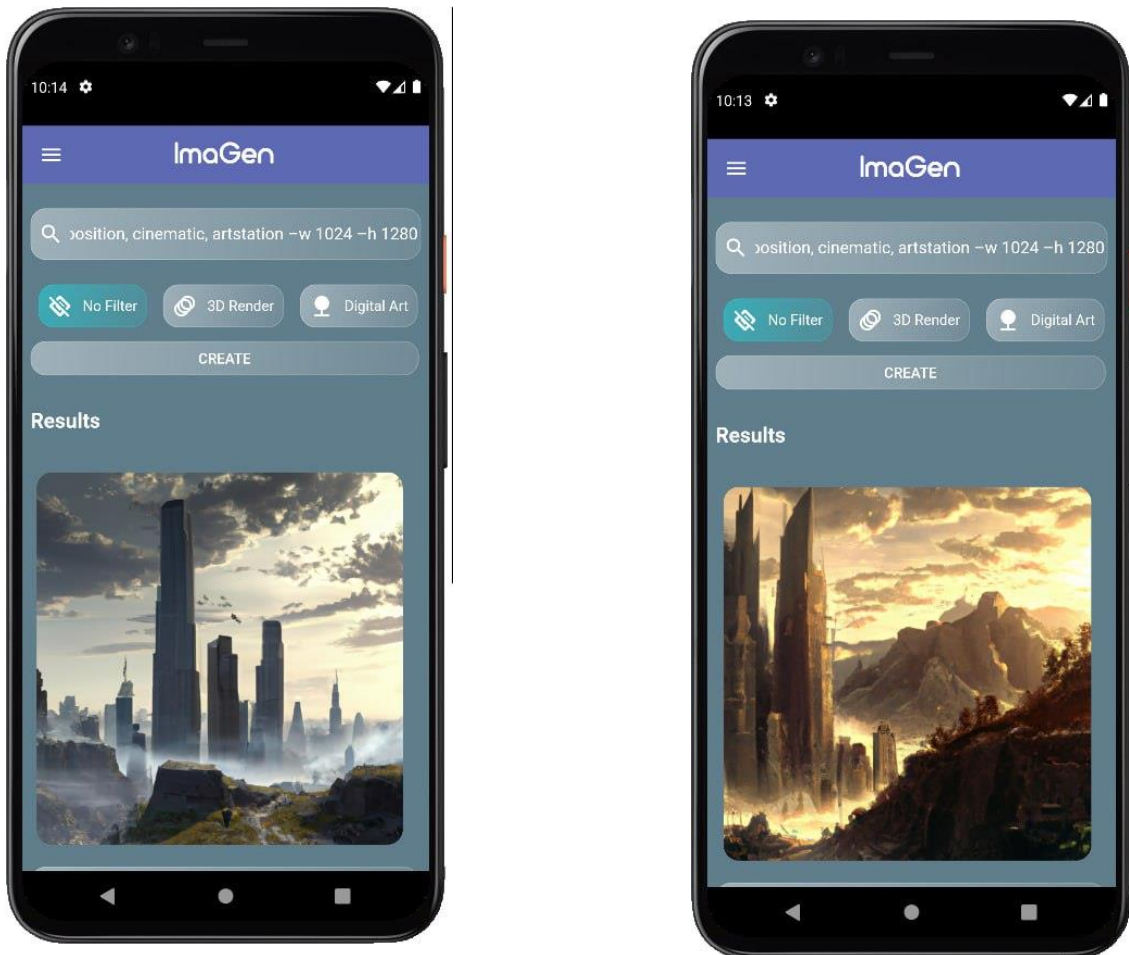


Figure 5.3: No Filter effect of the image

Text: The Legend of Zelda landscape atmospheric, hyper realistic, 8k, epic composition, cinematic, octane render.

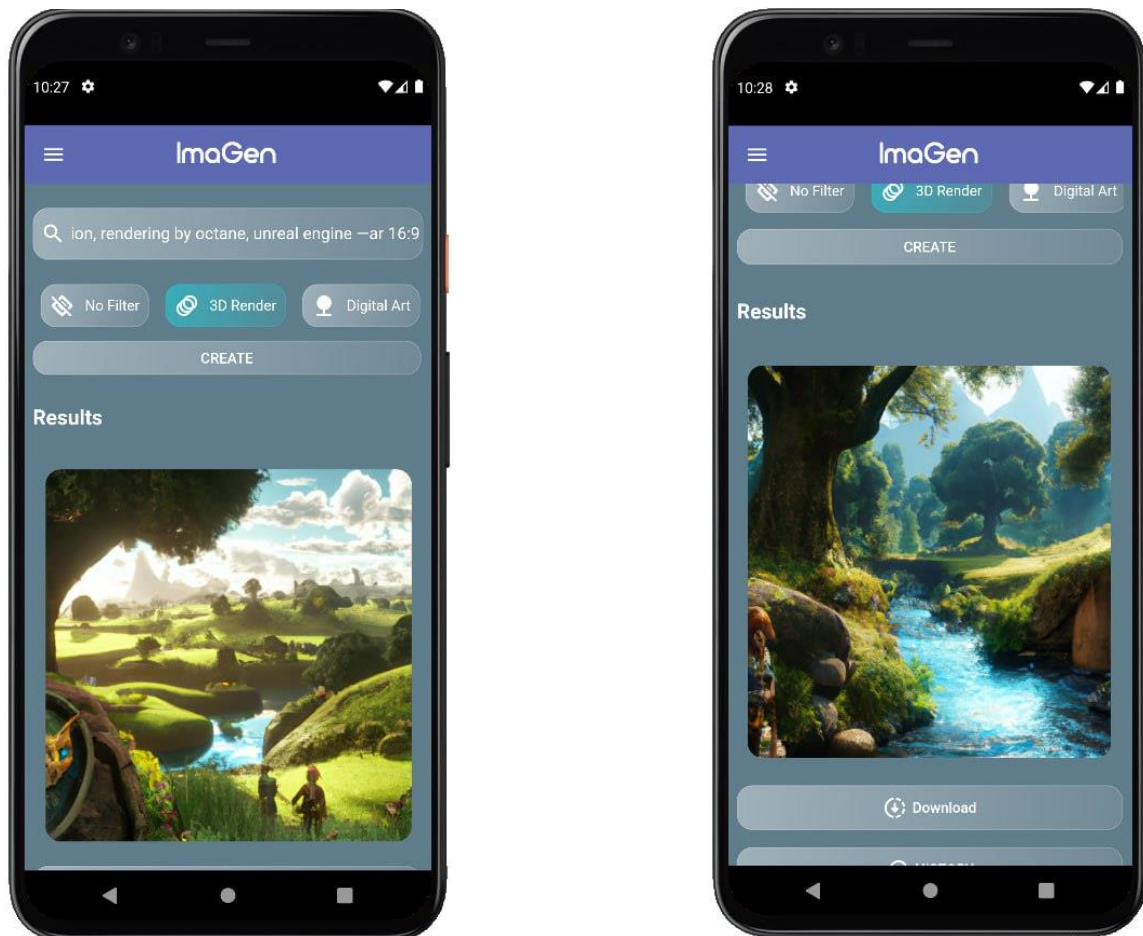


Figure 5.4: 3D Filter effect of the image

Text: a landscape by simon stalenhag of a very large realistic highly detailed imposing robotic mechanical cat.

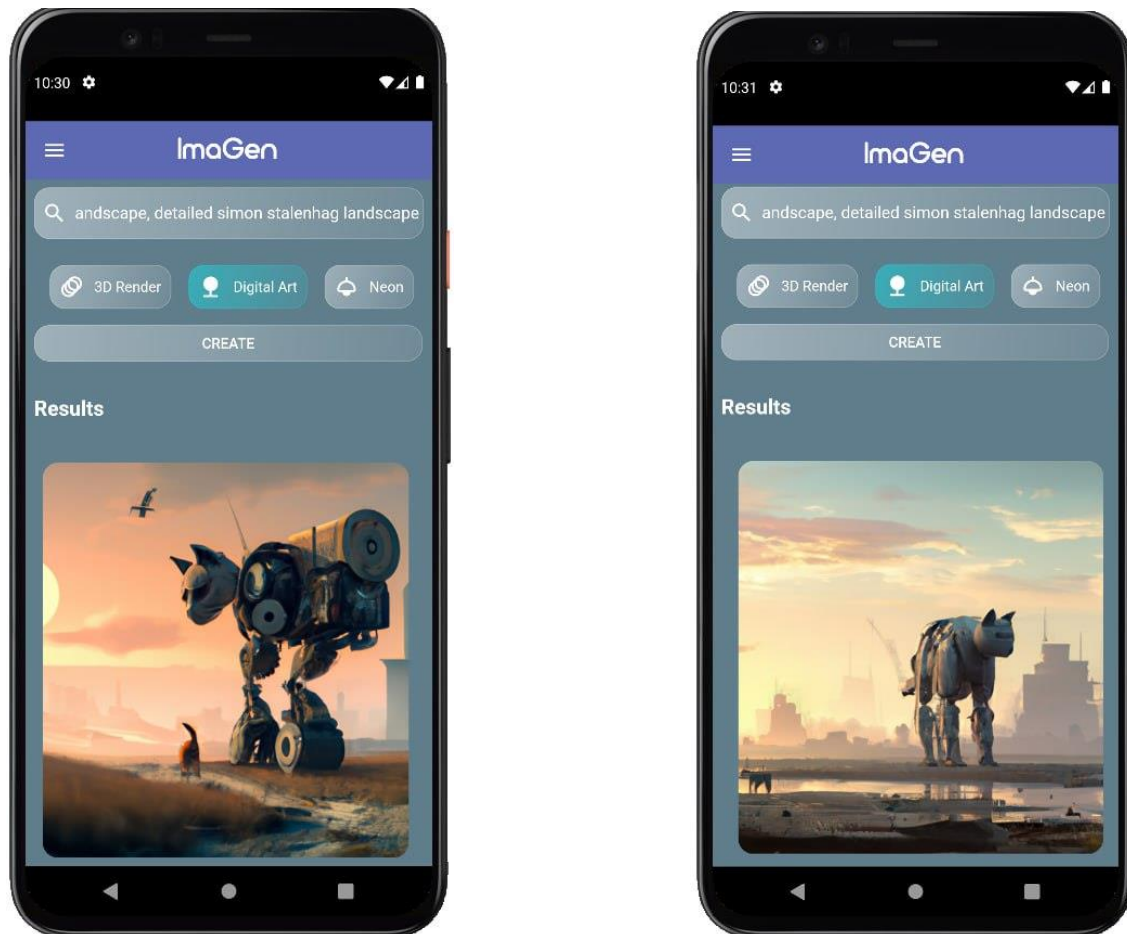


Figure 5.5: Digital art Filter effect of the image

Text: A grand city in the year 2100, atmospheric, hyper realistic, 8k, epic composition, cinematic, octane render, artstation landscape vista photography by Carr Clifton & Galen Rowell

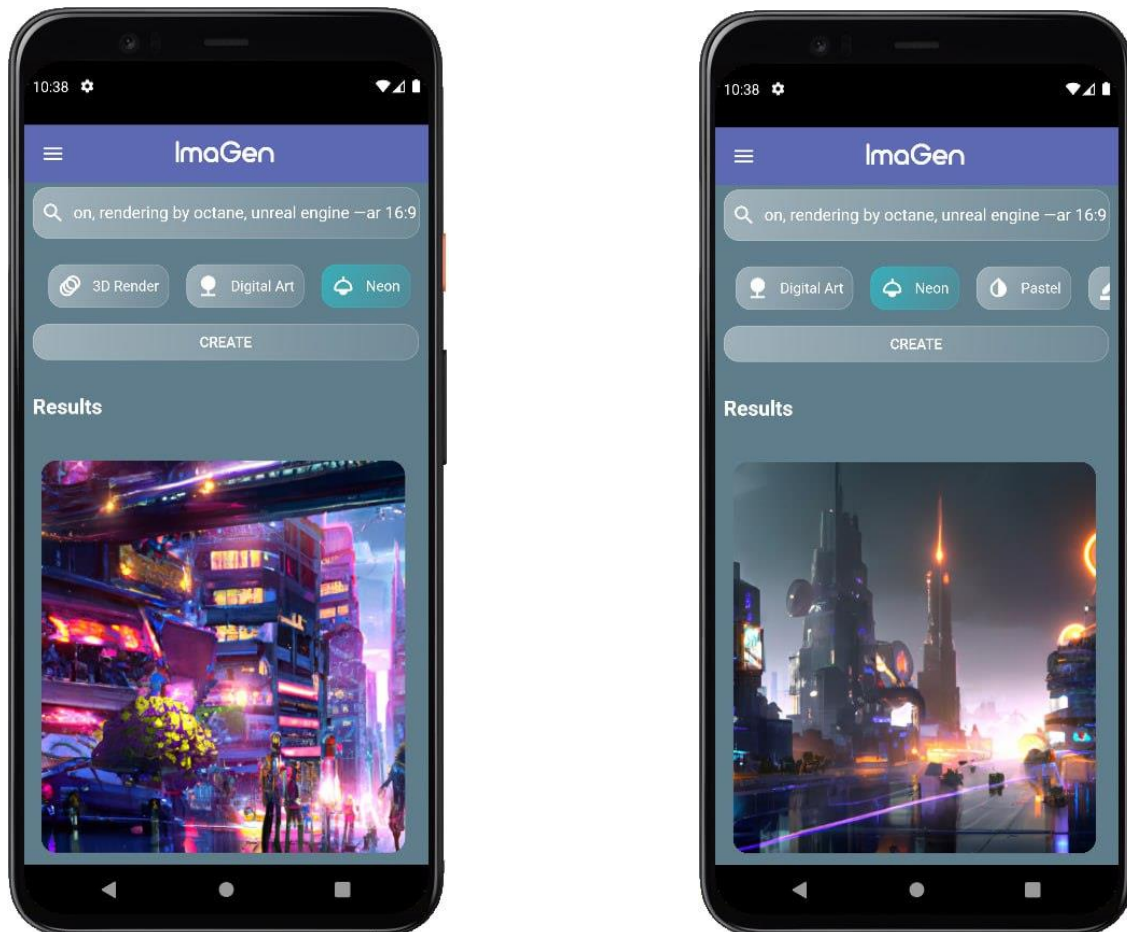


Figure 5.6: Neon Filter effect of the image

Text: walking on the starlight, dreamy ultra wide shot, atmospheric, hyper realistic, epic composition, cinematic, octane render, artstation landscape vista photography by Carr Clifton & Galen Rowell

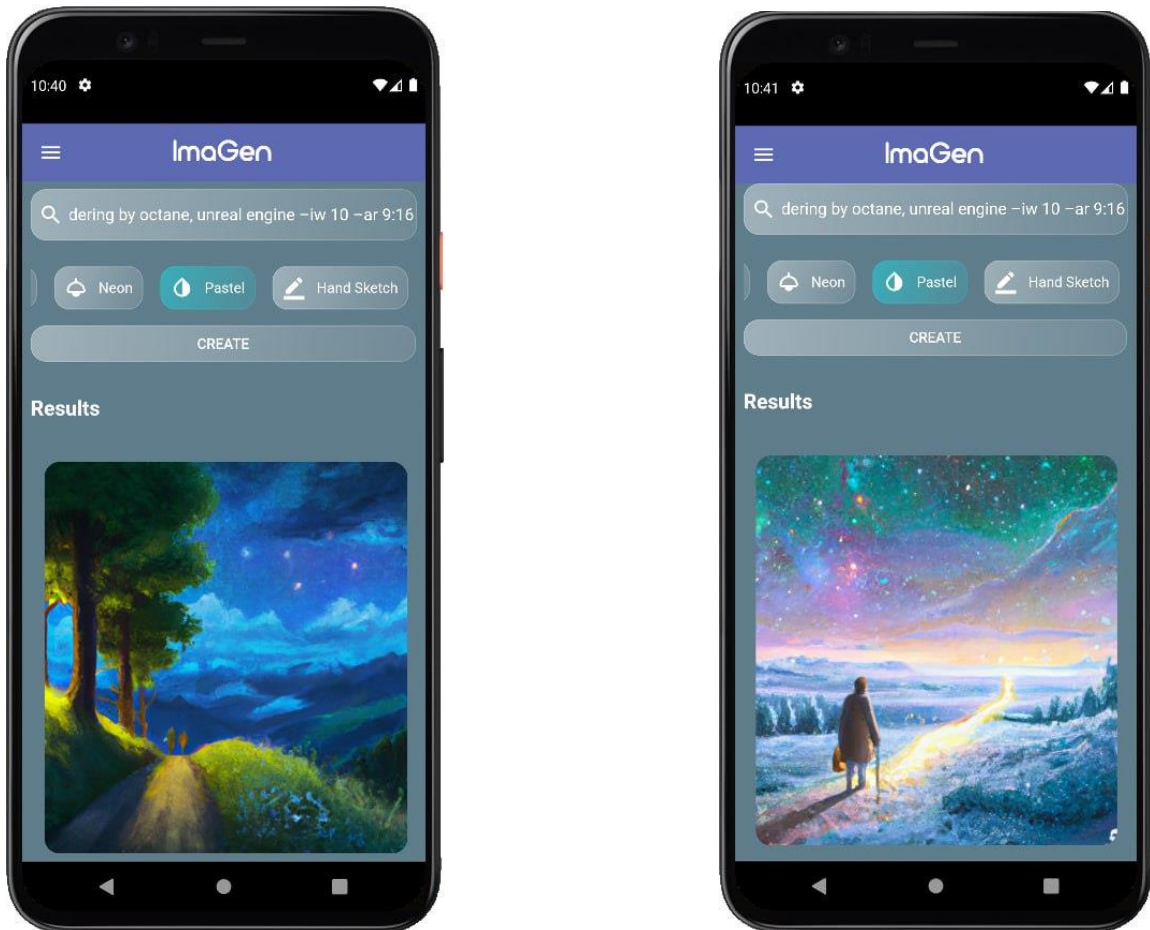


Figure 5.7: pastel Filter effect of the image

Text: futuristic nighttime cyberpunk New York City skyline landscape vista photography by Carr Clifton & Galen Rowell, 16K resolution, Landscape veduta photo by Dustin Lefevre & tdraw

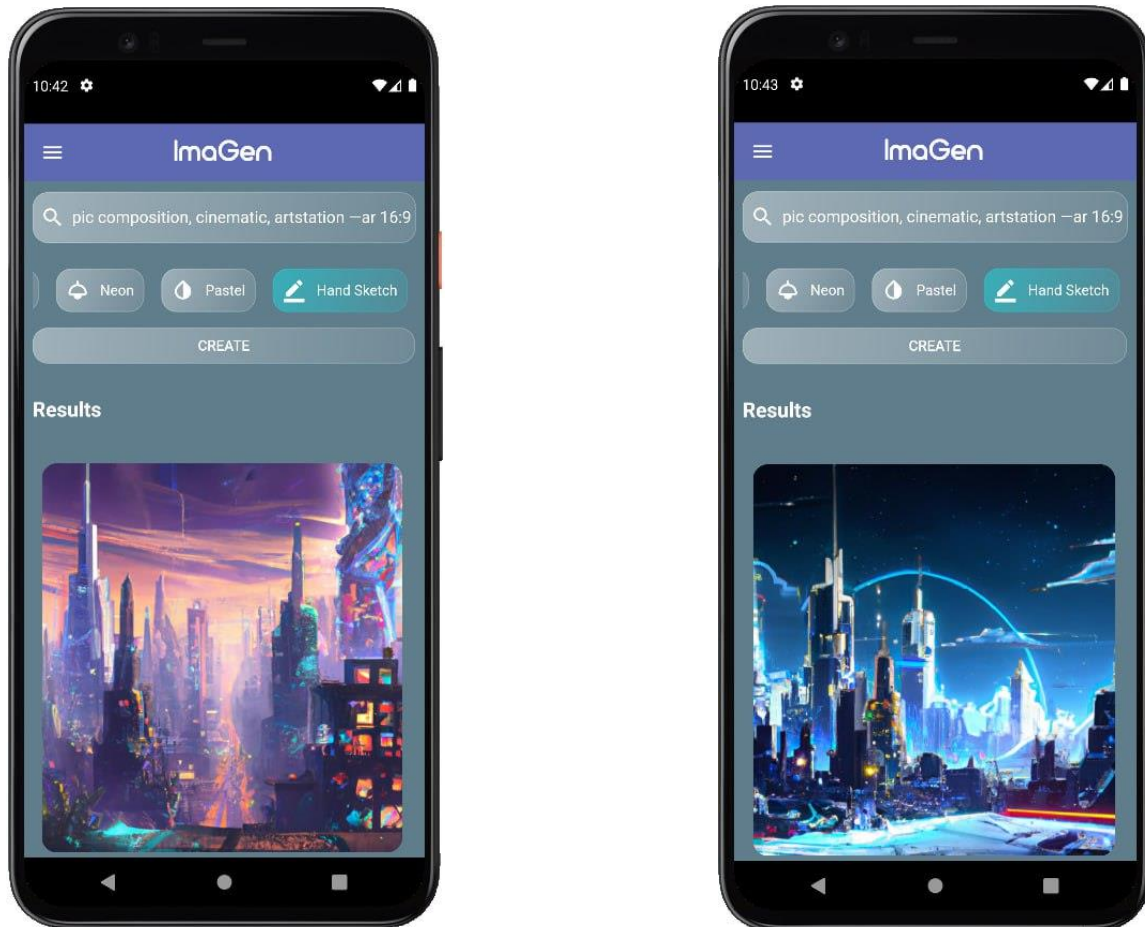


Figure 5.8: hand sketch Filter effect of the image

5.3 Implementation of Interaction

The success of every project hinges on interactive applications. Therefore, it is crucial for the project's success. Everyone in our nation who uses the internet expects applications to be user-friendly. If a program is overly complicated, users won't be interested. This is why the user-friendly application prioritized my involvement first. So I did my best to make this online application as user-friendly as I could.

- The user will see a search box to search for the image.
- There are six types of Filter buttons, which means you can select their images according to the filter.
- There is a Create button through which the image will be created.
- There is a history button through which users can find their searched images.

5.4 Testing Implementation

After successful running this app, I try to testing it in many ways. My whole app is tested by one module; user module. Given that some of example below:

Table 5.1: Testing Information and Date

Test Case	Test Input	Expected Outcome	Obtained Outcome	Result	Tested on
1.user Registration	Fill up form correctly	Successful Registration of user	Successful Registration	Pass	3/12/2022
2. User Registration	Your email part is not fulfill	Alert Please give your email	Form is incomplete, Please complete all field of form	Pass	9/12/2022

3. User Sign In	For log in please give your correct password	Alert Please give your correct password	Log In is error	Pass	13/12/2022
3.User Sign In	Please enter your email	Successful Registration of Customer	Alert message “Log In Successfully”	Pass	17/12/2022
4.prompt design	Enter your text here	Understand to model	Image Generate	Pass	21/11/2022
5.Filter design	Click the filter	Show result based on text	Alert message showed “Enter your address”	Pass	25/11/2020

5.5 Testing Result and Reports

We need to test the app to detect bugs. During testing, I encountered a lot of bugs. And practically all of the bugs I had have been fixed. I need to test my app more thoroughly before I can further simplify it.

CHAPTER 6

CONCLUSION AND FUTURE SCOPE

6.1 Discussion and Conclusion

Nowadays we need different types of images, may be a website design, wallpaper, advertisement, blog post etc. through this app we will provide realistic images to the user. This app will help the user to get his desired image in a very short time. Using this project As a result, users can save their time and money. From the future we will add new features according to user's requirements.

6.2 Goal

The main goal of our project is how to get a realistic and unique image from a text. Apart from this, the performance of the app, the user interface should be kept in mind so that it is useful for a user.

6.3 Limitations

Every software has some Limitations.

- No Customization, that means that users can't make images that are perfectly suited to their project brand, etc.
- Don't upload the image
- Subscription option is not added.
- No Website option

6.4 Scope for Further Developments

Any development project has no end to develop. So, I have also to develop this project in future. Here some of plans are given below:

- We can add image colorizer, Restorer.
- We can add object remover, background remover.

- We can add social media post categories, for example, Facebook post, Instagram post, story, Twitter post, you tube banner, you tube thumbnail etc.
- We can add subscription option.
- Use powerful algorithm for better performance.

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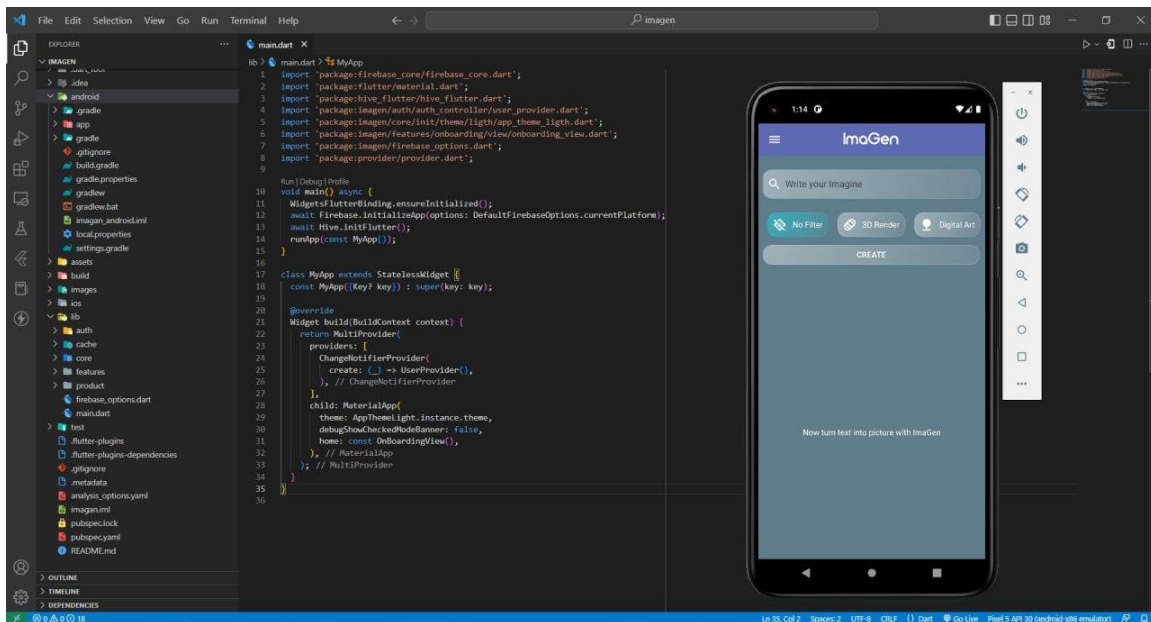
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APPENDIX

Appendix A: Project Reflection

we started working on this project in the summer of 2022. For this assignment, we performed a lot of. To use an ImaGen: AI Image Generator in Android and IOS platform, we've tried to learn about artificial intelligence techniques. we have gathered a lot of material from many papers and websites. then we decided to move on with this idea. Then we decided to utilize the CLIP model and diffusion model to generate images because it is a fundamental model that can generate images based on text. Trying to maintain it user-friendly is challenging. we create an intuitive user interface for this website. The user will easily understand this UI. So, we are hoping that will make things simpler for users. It will be extremely beneficial to users as more features and amenities are developed. ImaGen: AI Image Generator in Android and IOS platform software will therefore become popular to user.



PLAGIARISM

Plagiarism Report

ImaGen: AI Image Generator in Android and IOS platform

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